

Robustness Results in Linear-Quadratic Gaussian Based Multivariable Control Designs

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Abstract—The robustness of control systems with respect to model uncertainty is considered using simple frequency domain criteria. Available and new results are derived under a common framework in which the minimum singular value of the return difference transfer matrix is the key quantity. In particular, robustness results associated with multivariable control systems designed on the basis of linear-quadratic (LQ) and the linear-quadratic Gaussian (LQG) design methodologies are presented.

I. INTRODUCTION

THE importance of obtaining robustly stable feedback control systems has long been recognized by designers. Indeed, a principal reason for using feedback rather than open-loop control is the presence of model uncertainties

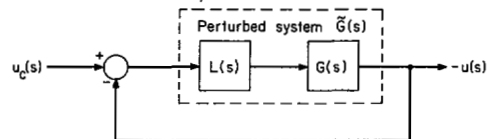


Fig. 1. Feedback system with multiplicative representation of uncertainty in $G(s)$.

minimum change in the model frequency response that leads to instability.

Commonly used measures of the closeness of an SISO feedback system to instability are its gain and phase margins. The margins are defined with reference to Fig. 1.

systems. However, the situation is quite different for multiple-loop systems. To quote again from the military specifications:

"In multiple-loop systems, variations shall be made with all gain and phase values in the feedback paths held at nominal values except for the path under investigation."

The fundamental difficulty with this approach is that it fails to check the effect of *simultaneous* gain and phase variations in several paths. Of course, real-world model uncertainty cannot be expected to nicely confine itself to a single loop of the system! In fact, for a flight control system, the dominant variation is due to the change in control surface effectiveness with dynamic pressure, which manifests itself as a change in the gains of the transfer functions from control surface deflections to the response variables of interest. The dynamic pressure variation is due to changes in aircraft altitude and speed, and clearly affects all loops simultaneously.

From this discussion, it is clear that a satisfactory notion of stability margins for a multivariable feedback system must be able to characterize the ability of the system to tolerate gain and phase variations in all its loops simultaneously. It is only very recently, in the context of studying the feedback properties of controllers derived using linear-quadratic Gaussian (LQG) techniques, that an appropriate formulation has emerged. The purpose of this paper is to develop the ideas of this formulation, and to derive in a relatively simple way the robustness properties of LQG controllers.

The paper begins in Section II by developing a characterization of the robustness of a multivariable feedback system in terms of the minimum singular value of its return difference transfer function matrix. This quantity is a natural one to consider as it directly generalizes the classical notion of the distance of the Nyquist locus of a single-input feedback system to the critical (-1) point.

NOTATION

(A, B, C)	realization of linear system specified by the time domain description $\dot{x} = Ax + Bu,$ $y = Cx,$
$G(s)$	loop-transfer matrix,
$\phi_{OL}(s)$	open-loop characteristic polynomial,
$\phi_{CL}(s)$	closed-loop characteristic polynomial,
$L(s)$	multiplicative perturbation of $G(s)$,
$\lambda(A)$	an eigenvalue of A ,
A^H	complex conjugate transpose of A ,
$\bar{\sigma}(A)$	maximum singular value of $A \triangleq \lambda_{\max}^{1/2}(A^H A),$
$\underline{\sigma}(A)$	minimum singular value of $A \triangleq \lambda_{\min}^{1/2}(A^H A),$
$\mathcal{N}(\Omega, f(s), \mathcal{C})$	number of clockwise encirclements of the point Ω by the locus of $f(s)$ as s traverses the closed contour $\mathcal{C}$ in the complex plane in a clockwise sense, Nyquist contour of radius R given in Fig. 2 with $(1/R)$ -radius indentations, segment of D_R for which $\text{Re}[s] \leq 0$.
D_R	
Ω_R	
SISO	single-input single-output,
MIMO	multiple-input multiple-output,
ORHP(CRHP)	open (closed) right half plane,
OLHP(CLHP)	open (closed) left half plane,
$A > B$	$A - B$ is positive definite matrix,
$A \geq B$	$A - B$ is positive semidefinite matrix,
$\Leftrightarrow$	if and only if,
$\exists$	there exists,
$\ni$	such that,
$\Rightarrow$	implies,
$\triangleq$	defined as,
LQ	linear-quadratic

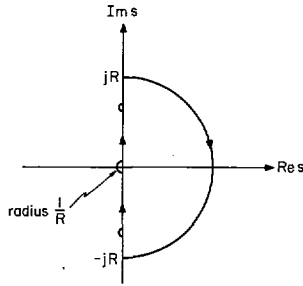


Fig. 2. Nyquist contour D_R which encloses all zeros of $\phi_{OL}(s)$ in the CRHP, avoiding imaginary zeros by indentations of radius of $1/R$.

ising the stability of the system. The analysis is based on the multivariable Nyquist theorem.

A. Multivariable Nyquist Theorem

The multivariable Nyquist theorem is derived from the relationship

$$\det(I + G(s)) = \frac{\phi_{CL}(s)}{\phi_{OL}(s)} \quad (2.2)$$

where

$$\phi_{OL}(s) = \det(sI - A) \quad (2.3)$$

$$\phi_{CL}(s) = \det(sI - A + BC) \quad (2.4)$$

and from the Principle of the Argument of complex variable theory. The following statement of the Nyquist theorem is a variation of a version given by Rosenbrock in [1].

Theorem 2.1: The system of Fig. 1 with $L(s) \equiv I$, is closed-loop asymptotically stable (in the sense that $\phi_{CL}(s)$ has no CRHP zeros) if and only if for all R sufficiently large

$$\mathcal{N}(0, \det[I + G(s)], D_R) = -P \quad (2.5)$$

or equivalently,

$$\mathcal{N}(-1, -1 + \det[I + G(s)], D_R) = -P \quad (2.6)$$

where D_R is the contour of Fig. 2 which encloses all CRHP zeros of $\phi_{OL}(s)$, P is the number of CRHP zeros of $\phi_{OL}(s)$ and where for convenience, we define $\mathcal{N}(\Omega, f(s), \mathcal{C}) \triangleq \frac{1}{2}$ when $f(s_0) = \Omega$, for some $s_0 \in \mathcal{C}$. ▲

Notice that no controllability or observability assumptions have been made. If $[A, B, C]$ is a nonminimal realization then pole-zero cancellations will occur when $G(s)$ is formed, eliminating uncontrollable or unobservable

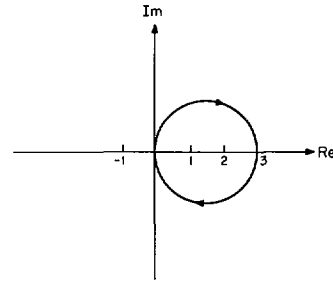


Fig. 3. Nyquist diagram of $2s+3/(s+1)^2$.

$G(s)$ is complicated, and cannot be easily depicted with a Nyquist, Bode, or related plot. This fact has motivated a considerable amount of research on synthesis methods, e.g., [2]–[5], which need not concern us here. Second, and this is the key observation, one cannot get a satisfactory notion of multivariable stability margins *directly* from the multivariable Nyquist theorem. The following extremely simple example illustrates this fact.

Example 2.1: Consider the linear system specified by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & b_{12} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad (2.7)$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (2.8)$$

which is illustrated in Fig. 4 by the nominal plant.

If the feedback compensation

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = - \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} u_{c1} \\ u_{c2} \end{bmatrix} \quad (2.9)$$

is used the closed-loop system is given by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & -b_{12} \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & b_{12} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_{c1} \\ u_{c2} \end{bmatrix}. \quad (2.10)$$

The eigenvalues of this system are $-2, -2$ and therefore it is stable. The return difference matrix $I + G(s)$ is given by

$$I + G(s) = \begin{bmatrix} \frac{s+2}{s+1} & \frac{b_{12}}{s+1} \\ 0 & \frac{s+2}{s+1} \end{bmatrix} \quad (2.11)$$

and thus

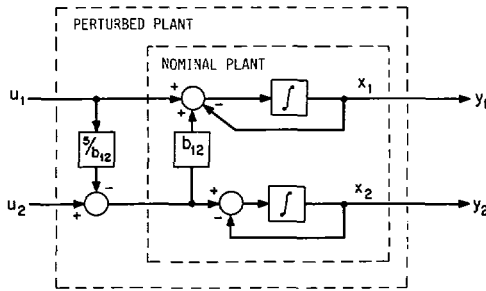


Fig. 4. Perturbation in nominal open-loop system which makes closed-loop system unstable.

margin of $\pm 106^\circ$ margins indicative of a highly robust system. Note, however, that the Nyquist diagram does not depend on the value of b_{12} and as b_{12} becomes large the closed-loop system is close to instability in the following sense. If the nominal open-loop system of Fig. 4 is perturbed slightly to obtain the perturbed system of Fig. 4, the closed-loop system obtained by negative identity feedback is unstable (as is easily verified). This situation cannot be detected by inspection of the multivariable Nyquist theorem.²

The difficulty we have uncovered can be explained as follows. A multivariable system will not be robust to modeling errors if its return difference transfer function matrix $I + G(j\omega)$ is nearly singular at some frequency ω_c , since then a small change in $G(j\omega_c)$ will make $I + G(j\omega_c)$ exactly singular. When this happens, $\det(I + G(j\omega_c)) = 0$ and the number of encirclements counted in the multivariable Nyquist theorem changes. In this example a small change in $I + G(j\omega_c)$ produces a large change in $\det(I + G(j\omega_c))$ showing that the near singularity of a matrix cannot always be detected in terms of its determinant. Instead, tests such as those developed in the remainder of this section must be employed.

B. Robustness Characterization

From the above example, we can see that the problem of determining characterizations of the robustness of a multivariable feedback system, i.e., its distance from instability, is of fundamental importance. The basic work in this area is due to Safonov [7], [8], who generalized an approach of Zames [10], [11]. Safonov's work heavily utilizes concepts of functional analysis, as is standard in the modern input-output formulation of stability theory.³ However, in the finite-dimensional linear time-invariant case considered in this paper, powerful robustness characterization can be derived more simply in terms of the multivariable Nyquist theorem.

The feedback system under consideration in the following theorem is shown in Fig. 1 where $G(s)$ is a

to $G(s)$ to account for uncertainty in the open-loop plant model.

We denote the perturbed system as $\tilde{G}(s)$ given by

$$\tilde{G}(s) = G(s)L(s) \quad (2.13)$$

which is assumed to have a state-space realization $(\tilde{A}, \tilde{B}, \tilde{C})$, open-loop characteristic polynomial $\tilde{\phi}_{OL}(s)$ given by

$$\tilde{\phi}_{OL}(s) = \det(sI - \tilde{A}) \quad (2.14)$$

and similarly a closed-loop characteristic polynomial $\tilde{\phi}_{CL}(s)$ given by

$$\tilde{\phi}_{CL}(s) = \det(sI - \tilde{A} + \tilde{B}\tilde{C}). \quad (2.15)$$

All the results of this section are based on the following theorem.

Theorem 2.2: The polynomial $\tilde{\phi}_{CL}(s)$ of (2.14) has no CRHP zeros if the following conditions hold:

a) $\phi_{OL}(s)$ and $\tilde{\phi}_{OL}(s)$ have the same number of CRHP zeros, (2.16)

b) if $\tilde{\phi}_{OL}(j\omega_0) = 0$ then $\phi_{OL}(j\omega_0) = 0$, (2.17)

c) $\phi_{CL}(s)$ has no CRHP zeros, (2.18)

d) $\det[I + (I - \epsilon)G(s) + \epsilon\tilde{G}(s)] \neq 0$, $s \in D_R$ for all R sufficiently large and for all ϵ on the interval $[0, 1]$. (2.19)

▲

The basic idea behind this theorem is that of continuously deforming the Nyquist diagram of the nominal system $G(s)$ to one corresponding to the Nyquist diagram of the perturbed system $\tilde{G}(s)$ without changing the number of encirclements of the critical point. If this can be done and the number of encirclements of the critical point required for $\tilde{G}(s)$ and $G(s)$ are the same, then no CRHP zeros of $\phi_{CL}(s)$ will result from this perturbation. Imbedding arguments of this type have been previously used, implicitly by Rosenbrock [1] and explicitly by Doyle [14], in connection with linear systems and in the more general context of nonlinear or multidimensional systems by DeCarlo, Saeks, and Murray [15]–[17], utilizing homotopy theory from algebraic topology.

The significance of Theorem 2.2 is that various multivariable robustness characterizations can be stated in terms of conditions that guarantee that (2.19) is satisfied. In checking condition (2.19), it is unnecessary to consider all $s \in D_R$ because $\bar{\sigma}[G(s)]$ and $\bar{\sigma}[\tilde{G}(s)] \rightarrow 0$ as $|s| \rightarrow \infty$. This is due to the assumption that their respective state-space realizations have no direct feedthrough from input to output so that there are more poles than zeros in both $\tilde{G}(s)$ and $G(s)$. It is therefore convenient to define Ω_R as

$$\Omega_R \triangleq \{s | s \in D_R \text{ and } \operatorname{Re}(s) \leq 0\} \quad (2.20)$$

which is the only part of the Nyquist contour on which

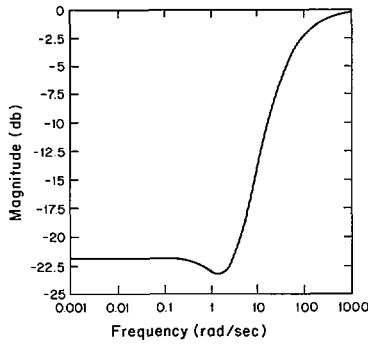


Fig. 6. Singular value plot of $\sigma(I+G(j\omega))$ for Example 2.1 when $b_{12}^- = 50$.

which give worst case bounds on the actual gain and phase margins. However, we know by inspection that each feedback loop in the system has an actual infinite upward gain margin and a -1 gain reduction margin which indicates that (2.50) and (2.51) are very conservative estimates of gain and phase margins. Nevertheless, they indicate a robustness problem which is exhibited by the very small crossfeed tolerance (see Corollary 2.2)

$$\bar{\sigma}(X(s)) < \min_{s \in \Omega_R} \underline{\sigma}(I+G(s)) \approx 0.071 = -23 \text{ dB}. \quad (2.52)$$

This again is a worst case bound on the allowable amount of crossfeed but corresponds precisely to the magnitude of the perturbation given previously in Fig. 4 to show nearness to instability. In general, the type of perturbation that the feedback system is most sensitive to at each s is given by

$$L(s) = [I - \underline{\sigma}(I+G(s))u(s)v^H(s)]^{-1} \quad (2.53)$$

where $u(s)$ and $v(s)$ are, respectively, the left and right singular vectors⁴ of $I+G(s)$ corresponding to $\underline{\sigma}(I+G(s))$. This is the perturbation which makes $I+G(s)L(s)$ exactly singular with minimum $\bar{\sigma}[(\tilde{G}^{-1}(s) - G^{-1}(s))G(s)]$.

Thus in Theorem 2.3 and its corollaries, if a global minimum of $\underline{\sigma}(I+G(s))$ occurs at $s_0 \in \Omega_R$ then there exists a perturbation $L(s)$ that satisfies

$$\bar{\sigma}(L^{-1}(s) - I) \leq \underline{\sigma}(I+G(s_0)) \quad \forall s \in \Omega_R \quad (2.54)$$

such that the closed-loop system is unstable. Note that $L(s)$ in (2.54) need correspond to $L(s)$ in (2.53) only at $s=s_0$ since $I+G(s)L(s)$ need only be singular at a single $s \in \Omega_R$ to induce instability of the closed-loop system.

III. LQ AND LQG STABILITY MARGINS

The subject of qualitative feedback properties of LQ control systems is not a new one. An early and fundamental paper by Kalman [21] detailed properties shared by all LO regulators in the single-input case. Kalman showed

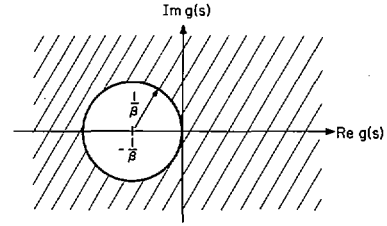


Fig. 7. Set of allowable values of $g(s)$ when $|1 + \beta g(s)| > 1$.

single-input LQ state feedback regulator satisfies the inequality

$$|1 + g(j\omega)| \geq 1, \quad \forall \omega. \quad (3.1)$$

This is both a classical condition for the reduction of sensitivity at the feedback input to the system (see, e.g., [19]) as well as necessary and sufficient for a (stable) state feedback regulator to be optimal with respect to some quadratic cost index. By inspection of the Nyquist diagram corresponding to (3.1), (Fig. 7 with $\beta=1$), it is straightforward to observe [22, pp. 70–76] that a SISO LQ state feedback regulator has a guaranteed infinite upward gain margin, at least a 50 percent gain reduction margin and also a guaranteed minimum phase margin of $\pm 60^\circ$.

Anderson [23] developed a multivariable version of condition (3.1) as a property of LQ state-feedback regulators; a similar generalized condition arises in sensitivity theory⁵ (see, e.g., Cruz and Perkins [24]). In the remainder of this paper, we will exploit the multivariable form of (3.1) together with the results of the preceding section to establish the stability margin properties of LQ and LQG optimal regulators.

A. Multivariable Kalman Inequality

We will need a precise statement of the multivariable version of condition (3.1) in the sequel, and this is provided by the following theorem. The proof is by straightforward manipulation of the algebraic Riccati equation and is included, for completeness, in the Appendix.

For convenience we will assume that in all remaining theorems and corollaries that the Nyquist contour D_R is chosen with R sufficiently large so that the theorems of Section II may be applied.

Theorem 3.1: If the matrix K satisfies the matrix algebraic Riccati equation

$$A^T K + KA + Q - KBR^{-1}B^T K = 0 \quad (3.2)$$

with $R > 0$ and $Q \geq 0$ then

$$(I+G(s))^H R (I+G(s)) = R + H(s) \quad (3.3)$$

$$G(s) = R^{-1}B^TK(Is-A)^{-1}B \quad (3.4)$$

$$H(s) = [(Is-A)^{-1}B]^H(Q+2\operatorname{Re}(s)K)[(Is-A)^{-1}B]. \quad (3.5)$$

Furthermore, if $Q > 0$, B has full rank and $K \geq 0$ then (3.3) implies that

$$(I+G(s))^HR(I+G(s)) > R, \quad s \in D_R. \quad (3.6)$$

Alternatively, if $\det(j\omega I - A) \neq 0 \forall \omega$ and $K \geq 0$ then (3.3) implies that

$$(I+G(s))^HR(I+G(s)) \geq R, \quad s \in D_R. \quad (3.7)$$

It is important to point out that this theorem uses $G^H(s)$ rather than $G^T(-s)$ as in [23]. These two quantities are the same when $s=j\omega$, but are different when $\operatorname{Re}(s) \neq 0$. This is the case when s is evaluated along the Nyquist D_R contour and this contour is indented along the imaginary axis. It is necessary to use $G^H(s)$ in order to apply the theorems of Section II. Note, however, that when $\det(j\omega I - A) \neq 0, \forall \omega$, that Ω_R is just the imaginary axis from $-jR$ to jR . In this case (3.7) could be written as

$$(I+G(j\omega))^HR(I+G(j\omega)) \geq R, \quad \forall \omega \quad (3.8)$$

which is the previously mentioned multivariable generalization of condition (3.1).

Of course, the Riccati equation (3.2) arises in connection with the LQ regulator problem defined by the optimi-

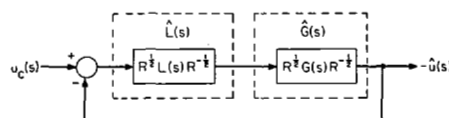


Fig. 8. Feedback system for stability margin derivation (compare Fig. 1).

then (3.7) (for example) can be rewritten in the form

$$(I+\hat{G}(s))^H(I+\hat{G}(s)) \geq I, \quad s \in D_R. \quad (3.12)$$

Equation (3.12) provides the bound

$$\sigma(I+\hat{G}(s)) \geq 1, \quad s \in D_R \quad (3.13)$$

on the minimum singular value of $I+\hat{G}(s)$.

To work with $\hat{G}(s)$ instead of $G(s)$, it is necessary to manipulate Fig. 1 into the equivalent (for stability analysis) form depicted in Fig. 8. Then using (3.6) and (3.7) together with Theorem 2.3 leads directly to the following result.

Theorem 3.2: The polynomial $\tilde{\phi}_{CL}(s)$ has no CRHP zeros provided the following conditions are satisfied:

a) conditions (2.16) and (2.17) hold (3.14)

b) $G(s)$ is specified by (3.4) where $K \geq 0$ satisfies (3.2) and $[A, B]$ is stabilizable, $[A, Q^{1/2}]$ is detectable and B has full rank (3.15)

c) with $\gamma(s) \triangleq \bar{\sigma}(R^{1/2}L^{-1}(s)R^{-1/2} - I)$ either of the following hold

i) $Q > 0$ and $\gamma(s) \leq 1, s \in \Omega_R$ (3.16)

ii) $\phi_{OL}(j\omega) \neq 0 \forall \omega$ and $\gamma(s) < 1, s \in \Omega_R$. (3.17)

Note that the condition $\bar{\sigma}(R^{1/2}L^{-1}(s)R^{-1/2} - I) \leq 1$ in

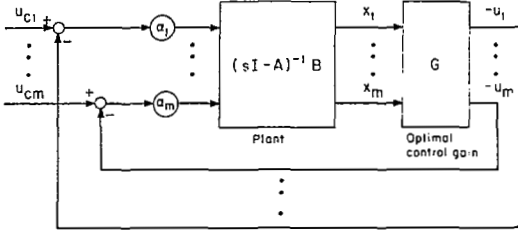


Fig. 9. Configuration for definition of multiloop LQ stability margins.

and also a guaranteed minimum phase margin (PM)

$$\text{PM} = \pm 60^\circ \quad (3.21)$$

if R is diagonal and either $Q > 0$ in (3.2) or $\phi_{OL}(j\omega) \neq 0, \forall \omega$.

If α_i represents a pure gain change or $\alpha_i = e^{j\phi_i}$ represents a pure phase change, then Fig. 9 illustrates the placement of the α_i in the feedback loop. Note that the interpretation of the gain and phase margins specified in (3.20) and (3.21) changes slightly depending on whether $Q > 0$ or $\phi_{OL}(j\omega) \neq 0, \forall \omega$ holds. If $Q > 0$, then the pure gain α_i must satisfy $\alpha_i \geq \frac{1}{2}$. Similarly, for $\alpha_i = e^{j\phi_i}$ we must have $|\phi_i| \leq 60^\circ$. If only $\phi_{OL}(j\omega) \neq 0, \forall \omega$ holds then pure gains α_i must satisfy $\alpha_i > \frac{1}{2}$ while pure phase factors $\alpha_i = e^{j\phi_i}$ must satisfy $|\phi_i| < 60^\circ$.

Results related to Corollary 3.1 have been derived by various authors [26]–[29]; but the definitive treatment including the multivariable phase margin result is due to Safonov and Athans [25]. The approach of this paper, based on relatively simple frequency domain arguments, is new.

If R is not diagonal then the guarantees of Corollary 3.1 do not apply. The following example illustrates that the gain margins may become arbitrarily small.

Example 3.1: Consider the LQ regulator specified by (3.15) with

$$(A, B, Q^{1/2}) = \left(I_2, \begin{bmatrix} 1 & \beta \\ 0 & 1 \end{bmatrix}, I_2 \right) \quad (3.22)$$

where I_2 is the 2×2 identity matrix and $R > 0$ is a nondiagonal control-weighting matrix given by

$$R = B^T [K^{-2} + 2K^{-1}]^{-1} B \quad (3.23)$$

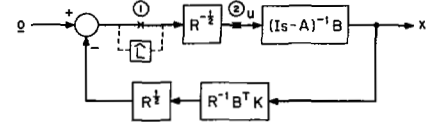
where $K > 0$ is arbitrary. By selection of R in (3.23), K satisfies (3.2). Now let the multiplicative perturbation $L(s)$ be given by the constant matrix L where

$$L = \begin{bmatrix} 1 & 0 \\ 0 & 1 + \epsilon \end{bmatrix} \quad (3.24)$$

and $\epsilon \neq 0$ is arbitrary. The zeros of $\tilde{\phi}_{CL}(s)$ are the eigenvalues of the perturbed closed-loop system matrix $\tilde{A}_{CL}$ where

$$\tilde{A}_{CL} = A - BLB^{-1}(2I + K^{-1}) \quad (3.25)$$

or

Fig. 10. LQ regulator with margins guaranteed at point ① for an $R > 0$ and at both ① and ② for diagonal $R > 0$.

$$\tilde{A}_{CL} = - \begin{bmatrix} (p_1 + 2) + \beta \epsilon p_2 - 1 & p_2 + \beta \epsilon (p_3 + 2) \\ (1 + \epsilon) p_2 & (1 + \epsilon)(p_3 + 2) - 1 \end{bmatrix} \quad (3.26)$$

where we have let K^{-1} be denoted by

$$K^{-1} = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix}. \quad (3.27)$$

For $\tilde{A}_{CL}$ to have no CRHP eigenvalues it is necessary for $\text{tr} \tilde{A}_{CL} < 0$. However, by inspection of (3.26), if $p_2 \neq 0$ then for any $\epsilon \neq 0$ there exists a β that will make $\text{tr} \tilde{A}_{CL} > 0$ and therefore for arbitrarily small ϵ , the perturbed closed-loop system will be unstable.

The basic problem exposed here is that the margins are really guaranteed at a different point in the loop than where we would like. This is illustrated in Fig. 10 where the perturbation $\hat{L}(s)$ is inserted at point ①. When $\hat{L}(s)$ is diagonal, as when calculating gain and phase margins, and R is also diagonal then $R^{-1/2}$ and $\hat{L}(s)$ commute and points ① and ② have identical guaranteed gain and phase margins. Point ② is where it is important to have margins (i.e., at the input to the physical plant), not inside the compensator at point ①.

Returning to Example 2.1 of the previous section once more, an LQ feedback control law is given that has the same closed-loop poles as before, but avoids the near instability associated with the negative identity feedback.

Example 3.2: With $b_{12} = 50$ in (2.10) as in the plot of $\sigma(I + G(j\omega))$, an LQ design using $R = 1$ and

$$Q = 3 \begin{bmatrix} 2601 & -50 \\ -50 & 1 \end{bmatrix} \quad (3.28)$$

gives a feedback gain of

$$R^{-1} B^T K = \begin{bmatrix} 1 & -50 \\ 0 & 1 \end{bmatrix} \quad (3.29)$$

and a closed-loop system matrix A_{CL} of

$$A_{CL} = A - BR^{-1} B^T K = -2I. \quad (3.30)$$

This makes $I + G(s)$

$$I + G(s) = \begin{bmatrix} \frac{s+2}{s+1} & 0 \\ 0 & \frac{s+2}{s+1} \end{bmatrix} \quad (3.31)$$

and thus

$$\sigma(I + G(j\omega)) = \left[\frac{\omega^2 + 4}{\omega^2 + 1} \right]^{1/2} \geq 1. \quad (3.32)$$

As one might expect the ability of LQ regulators to tolerate crossfeed perturbations is also affected by the choice of the control weighting matrix R . This is made precise in the following corollary.

Corollary 3.2: The LQ regulator with loop transfer matrix $G(s)$ satisfying (3.15) will tolerate (i.e., $\tilde{\phi}_{CL}(s)$ will have no CRHP zeros) a crossfeed perturbation of the form (2.42) satisfying (2.16) and (2.17) provided

$$\bar{\sigma}^2(X(s)) < \min \left\{ \frac{\lambda_{\min}(R_1)}{\lambda_{\max}(R_2)}, \frac{\lambda_{\min}(R_2)}{\lambda_{\max}(R_1)} \right\}, \quad s \in \Omega_R \quad (3.33)$$

Theorem 3.3: For $G(s)$ of the form of (3.4), $\tilde{\phi}_{CL}(s)$ has no CRHP zeros if the following conditions hold:

a) $\tilde{\phi}_{OL}(s)$ has no CRHP zeros (3.38)

b) $K \geq 0$ satisfies (3.36) with $Q \geq 0$, $R > 0$ and $[A, Q^{1/2}]$ detectable, and B has full rank (3.39)

c) either of the following holds

i) $Q > 0$ and $RL(s) + L^H(s)R \geq 0, s \in \Omega_R$ (3.40)

ii) $RL(s) + L^H(s)R > 0, s \in \Omega_R$. (3.41)

▲

Corollary 3.3: For $G(s)$ as in Theorem 3.3 with R diagonal the guaranteed gain and phase margins are given by

$$GM = 0, \infty \quad (3.42)$$

Proof: Similar to Corollary 2.3

LQ optimal feedback law associated with (3.45) is given by

$$u(t) = -\beta R^{-1} B^T K x(t). \quad (3.46)$$

Instead of (3.46) we will use $u(t) = -R^{-1} B^T K x(t)$ as in (3.11). Thus depending on whether $\beta > 1$ or $\beta < 1$ we are merely decreasing or increasing, respectively, the optimal feedback gain by a scalar factor of $1/\beta$. Also with $G(s)$ given by (3.4) the standard LQ loop transfer matrix is simply $\beta G(s)$. From Theorem 3.1 we know that if $Q > 0$

$$[I + G(s)]^H \frac{1}{\beta} R [I + \beta G(s)] > \frac{1}{\beta} R, \quad s \in \Omega_R \quad (3.47)$$

which in the SISO case becomes

$$\left| \frac{1}{\beta} + g(s) \right| > \frac{1}{\beta}, \quad s \in \Omega_R \quad (3.48)$$

and is illustrated in Fig. 7. To obtain bounds on $L(s)$ to ensure stability we merely work with $1/\beta L(s)$ and $\beta G(s)$ and apply Theorem 3.2 for the standard LQ regulator problem. Doing this we obtain, in the SISO case, the inequality

$$|\beta l^{-1}(s) - 1| < 1, \quad s \in \Omega_R \quad (3.49)$$

illustrated in Fig. 11. Note that from Fig. 7 that the critical "1" point is no longer contained inside the circle

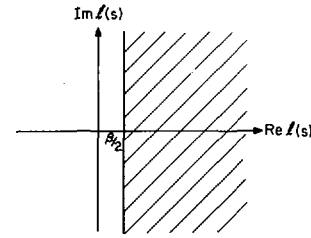


Fig. 11. Set of allowable values of $l(s)$ when $|\beta l^{-1}(s) - 1| < 1$ and $0 < \beta < 2$.

nondivergent Kalman filter under variations in the nominal model parameters of the plant whose state is to be estimated. To make the precise connection between the regulator and filter problems, consider the linear system

$$\dot{x}(t) = A x(t) + \xi(t) \quad (3.52)$$

$$y(t) = C x(t) + \theta(t) \quad (3.53)$$

where $\xi(t)$ and $\theta(t)$ are zero mean white noise sources with spectral intensity matrices Ξ and Θ , respectively. We wish to estimate $x(t)$ given $y(\tau)$, $-\infty \leq \tau \leq t$, such that the mean square error is minimized. Under the assumption the $[A, C]$ is detectable, it is well-known that the state estimate is specified by

$$\dot{\hat{x}}(t) = A \hat{x}(t) + \Sigma C^T \Theta^{-1} v(t) \quad (3.54)$$

$$v(t) = y(t) - C \hat{x}(t) \quad (3.55)$$

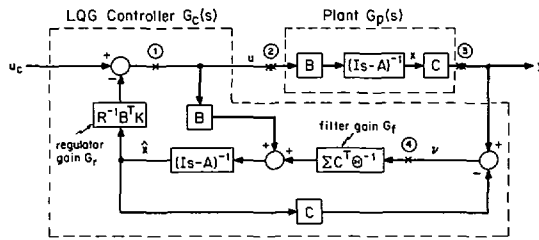


Fig. 12. LQG control system.

filter. Furthermore, they have considered the robustness properties of a nonlinear LQG control system formed by the cascade of a constant gain extended Kalman filter and the LQ state feedback gain. The LQ state feedback gain and the constant gain of the extended Kalman filter are computed from the linearized model parameters. However, the extended Kalman filter must have the true non-linear model of the plant. In the completely linear case the

loop plant $G_p(s) \triangleq C(Is-A)^{-1}B$ if the following conditions hold.

a) The perturbed open-loop plant $\tilde{G}_p(s) \triangleq \tilde{C}(Is-\tilde{A})^{-1}\tilde{B}$ is such that the $\det(sI-\tilde{A})$ and $\det(sI-A)$ have the same number of CRHP zeros and if $\det(j\omega_0 I-\tilde{A})=0$ then $\det(j\omega_0 I-A)=0$. (3.69)

b) $[A, B]$ is stabilizable, $Q>0$, $R>0$ and $K\geq 0$ satisfies (3.2) and B has full rank. (3.70)

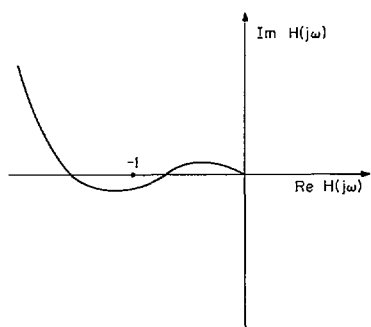
c) $\tilde{G}_p(s) = G_p(s)L(s) = N(s)G_p(s)$ and either (3.71)

$$\bar{\sigma}(R^{1/2}L^{-1}(s)R^{-1/2}-I) \leq 1 \quad (3.72)$$

or

$$\bar{\sigma}(\Theta^{-1/2}N^{-1}(s)\Theta^{1/2}-I) \leq 1 \quad (3.73)$$

hold for all $s \in \Omega_\infty$.



The quantity k is the minimum of the condition numbers of $G_p(s)$ and $G_c(s)$ with respect to inversion. The proof is accomplished by a simple calculation and is omitted. From (3.77) we conclude that if k is close to unity then approximately the same robustness guarantees will apply at both input and output. Note that we have no control over $G_p(s)$ so that if $G_p(s)$ is nearly singular we must design our compensator so that $\bar{\sigma}(G_c(s)) \cong \underline{\sigma}(G_c(s))$. On the other hand, if our plant is well-conditioned with respect to inversion our compensator $G_c(s)$ need not be so severely constrained, allowing more flexibility in achieving

generally valid point that stability margins are completely useless in themselves without further information about the size of the expected perturbations in the nominal system.

B. Conclusions and Future Research

Although the singular value approach is useful in detecting the near instability of a control system, it is sometimes unnecessarily conservative. This is due to the fact that some of the small perturbations that would theoretically destabilize the closed-loop system will never occur in the physical system. Nevertheless they are still detected by a small $\sigma(I+G(s))$. One direction for further research is characterization of the robustness of a feedback system in which perturbations in certain directions are ruled out as impossibilities on physical grounds. This points out the fundamental problem of obtaining the characterization of the uncertainty associated with a given model. It seems

where P is the number of CRHP zeros of $\phi_{OL}(s)$ and, by (2.16), also of $\tilde{\phi}_{OL}(s)$. Clearly $\det[I+(1-\epsilon)G(s)+\epsilon\tilde{G}(s)]$ is a continuous function of ϵ for all $s \in D_R$ and from (A.1)

$$\mathcal{N}(0, \det[I+(1-\epsilon)G(s)+\epsilon\tilde{G}(s)], D_R)|_{\epsilon=0} = -P. \quad (A.2)$$

Now suppose that ϵ is varied continuously from 0 to 1 and that $\mathcal{N}(0, \det[I+(1-\epsilon)G(s)+\epsilon\tilde{G}(s)], D_R)$ does not remain constant at $-P$. From the Principle of the Argument we know that for some ϵ on $[0, 1]$ the number of zeros minus the number of poles of $\det[I+(1-\epsilon)G(s)+\epsilon\tilde{G}(s)]$ enclosed in D_R must change. However, since the poles and zeros are also continuous functions of ϵ , it must be that for some ϵ_0 on $[0, 1]$ that they lie on D_R and thus

$$\det[I+(1-\epsilon_0)G(s)+\epsilon_0\tilde{G}(s)] = 0 \text{ or } \infty. \quad (A.3)$$

Defining $z = Lx$, (A.11) implies $z \neq 0$ and

$$z = -LGz. \quad (\text{A.12})$$

Condition (A.6) and (A.12) imply

$$z^H Gz + z^H G^H z = -z^H G^H [L + L^H] Gz \geq 0 \quad (\text{A.13})$$

and since $Gz \neq 0$ a contradiction to (A.6) and (A.10) is obtained. The proof for c) is analogous to that of b).
Q.E.D.

Lemma A.2: If L is a square matrix and $P(\epsilon) = (1 - \epsilon)I + \epsilon L$, then for all ϵ on $[0, 1]$ the following implications hold:

$$\text{a) } \bar{\sigma}(L^{-1} - I) < \alpha \leq 1 \Rightarrow \bar{\sigma}(P^{-1}(\epsilon) - I) < \alpha \quad (\text{A.14})$$

$$\text{b) } \bar{\sigma}(L^{-1} - I) < \alpha \text{ and } L + L^H \geq 0 \Rightarrow \bar{\sigma}(P^{-1}(\epsilon) - I) < \alpha \quad (\text{A.15})$$

$$\text{c) } \bar{\sigma}(L^{-1} - I) < \alpha \text{ and } 4(\alpha^2 - 1)\sigma^2(L - I) > \alpha^2 \bar{\sigma}^2(L + L^H - 2I) \Rightarrow \bar{\sigma}(P^{-1}(\epsilon) - I) < \alpha \quad (\text{A.16})$$

$$\text{d) } L + L^H > 0 \Rightarrow P(\epsilon) + P^H(\epsilon) > 0. \quad (\text{A.17})$$

Proof: For a) and b) rewrite $\bar{\sigma}(P^{-1}(\epsilon) - I) < \alpha$ as

$$\alpha^2 P^H(\epsilon) P(\epsilon) - (P(\epsilon) - I)^H (P(\epsilon) - I) > 0. \quad (\text{A.18})$$

Expanding the left-hand side of (A.18) gives

$$\begin{aligned} & \alpha^2 P^H(\epsilon) P(\epsilon) - (P(\epsilon) - I)^H (P(\epsilon) - I) \\ &= \epsilon^2 [\alpha^2 L^H L - (L - I)^H (L - I)] \\ & \quad + \alpha^2 (1 - \epsilon) [(1 - \epsilon)I + \epsilon(L + L^H)]. \end{aligned} \quad (\text{A.19})$$

By (A.14) or (A.15) the first term on the right-hand side of (A.19) is assumed to be positive definite. Now with $\alpha \leq 1$, (A.14) also gives

$$L^H + L > I + (1 - \alpha^2)L^H L > 0 \quad (\text{A.20})$$

so that (A.14) or (A.15) guarantee $L + L^H \geq 0$ which makes the second term on the right-hand side of (A.19) positive semidefinite. Thus (A.18) is true for all ϵ on $[0, 1]$ which is equivalent to $\bar{\sigma}(P^{-1}(\epsilon) - I) < \alpha$. To prove c) rewrite (A.19) as

$$\begin{aligned} & \alpha^2 P^H(\epsilon) P(\epsilon) - (P(\epsilon) - I)^H (P(\epsilon) - I) \\ &= (\alpha^2 - 1)\epsilon^2 (L - I)^H (L - I) + \alpha^2 \epsilon [L + L^H - 2I] + \alpha^2 I. \end{aligned} \quad (\text{A.21})$$

For the right-hand side of (A.21) to be positive definite, it is sufficient to show that $f(\epsilon) > 0$ where $f(\epsilon)$ is given by

$$f(\epsilon) = \beta^2 \epsilon^2 - 2\phi \epsilon + \alpha^2 \quad (\text{A.22})$$

and where

$$\beta^2 = (\alpha^2 - 1)\sigma^2(L - I) \quad (\text{A.23})$$

$$\phi = \alpha^2 \bar{\sigma} \left(\frac{L + L^H}{2} - I \right). \quad (\text{A.24})$$

Differentiating $f(\epsilon)$ to find an ϵ^* such that $f(\epsilon^*)$ is minimum results in

$$\epsilon^* = \frac{\phi}{\beta^2} \quad (\text{A.25})$$

and

$$f(\epsilon^*) = \alpha^2 - \frac{\phi^2}{\beta^2}. \quad (\text{A.26})$$

Thus $f(\epsilon^*)$ will be positive if

$$\alpha^2 \beta^2 > \phi^2 \quad (\text{A.27})$$

or

$$4(\alpha^2 - 1)\sigma^2(L - I) > \alpha^2 \bar{\sigma}^2(L + L^H - 2I) \quad (\text{A.28})$$

which proves c).

Implication d) follows trivially by direct substitution.
Q.E.D.

Proof of Theorems 2.3 and 2.4: Conditions (2.21) and (2.44) ensure that only condition (2.19) of Theorem 2.2 need be verified. Since $\bar{\sigma}[G(s)]$ and $\bar{\sigma}[\tilde{G}(s)] \rightarrow 0$ as $|s| \rightarrow \infty$, $\det[I + (1 - \epsilon)G(s) + \epsilon\tilde{G}(s)] \neq 0$ for all $|s| = R$ for R sufficiently large. Thus (2.19) need only be checked for $s \in \Omega_R$. Lemmas A.1 and A.2 applied at every $s \in \Omega_R$, guarantee that (2.19) holds for $s \in \Omega_R$ and thus for $s \in D_R$. The conditions of Theorem 2.2 are satisfied and thus $\phi_{CL}(s)$ has no CRHP zeros.
Q.E.D.

Proof of Corollaries 2.1 and 2.3: In Theorems 2.3 and 2.4 take $L(s)$ to be a diagonal matrix given by

$$L(s) = \text{diag}[l_1(s), l_2(s), \dots, l_n(s)] \quad (\text{A.29})$$

which simplifies conditions (2.22) and (2.23) to

$$|l_i^{-1}(s) - 1| < \alpha_0, \quad \forall i \quad (\text{A.30})$$

and condition (2.46) to

$$\text{Re}[l_i(s)] > 0, \quad \forall i. \quad (\text{A.31})$$

To obtain gain margins, let

$$l_i(s) = l_i, \quad l_i \text{ real}$$

then (A.30) becomes

$$\frac{1}{1 + \alpha_0} < l_i < \frac{1}{1 - \alpha_0} \quad (\text{A.32})$$

and (A.31) becomes

$$l_i > 0. \quad (\text{A.33})$$

Similarly, to obtain phase margins, let

$$l_i(s) = e^{j\phi_i(s)}, \quad \phi_i(s) \text{ real},$$

then (A.30) becomes

$$1 - \frac{\alpha_0^2}{2} < \cos \phi_i(s) \quad (\text{A.34})$$

or

$$|\phi_i(s)| < \cos^{-1} \left[1 - \frac{\alpha_0^2}{2} \right] \quad (\text{A.35})$$

while (A.31) becomes

$$\cos \phi_i(s) > 0 \quad (\text{A.36})$$

or

$$|\phi_i(s)| < \pm 90^\circ. \quad \text{Q.E.D.} \quad (\text{A.37})$$

Proof of Corollaries 2.2 and 2.4: Only conditions (2.22) and (2.46) need verification. With $L(s)$ given by (2.42) we have

$$L^{-1}(s) - I = \begin{bmatrix} 0 & -X(s) \\ 0 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & 0 \\ -X(s) & 0 \end{bmatrix} \quad (\text{A.38})$$

and thus

$$\bar{\sigma}(L^{-1}(s) - I) = \bar{\sigma}(X(s)) < \alpha_0 \leq \underline{\sigma}(I + G(s)). \quad (\text{A.39})$$

Also,

$$L(s) + L^H(s) = \begin{bmatrix} 2I & X(s) \\ X^H(s) & 2I \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 2I & X^H(s) \\ X(s) & 2I \end{bmatrix} \quad (\text{A.40})$$

so that

$$\begin{aligned} \lambda_{\min}(L(s) + L^H(s)) &\geq 2 - \lambda_{\max} \left(\begin{bmatrix} 0 & X(s) \\ X^H(s) & 0 \end{bmatrix} \right) \\ &\geq 2 - \bar{\sigma} \left(\begin{bmatrix} 0 & X(s) \\ X^H(s) & 0 \end{bmatrix} \right) \\ &= 2 - \bar{\sigma}(X(s)) > 0. \end{aligned} \quad (\text{A.41})$$

Q.E.D.

Proof of Theorem 3.1: Direct manipulation of (3.2) gives

$$\begin{aligned} (s^*I - A^T)K + K(sI - A) + KBR^{-1}B^TK \\ = (Q + 2\operatorname{Re}(s)K) \end{aligned} \quad (\text{A.42})$$

where s^* denotes the complex conjugate of s . Premultiplying and postmultiplying (A.42) by $[(sI - A)^{-1}B]^H$ and $[(sI - A)^{-1}B]$ respectively, we obtain

$\hat{G}(s) \triangleq R^{1/2}G(s)R^{-1/2}$, we see that $G(s)$ has a state-space realization $(A, BR^{-1/2}, R^{-1/2}B^TK)$ and thus its open- and closed-loop characteristic polynomials $\hat{\phi}_{OL}(s)$ and $\hat{\phi}_{CL}(s)$ are identical to those of $(A, B, R^{-1}B^TK)$. Thus any assumptions about $\phi_{OL}(s)$ and $\phi_{CL}(s)$ obviously apply to $\hat{\phi}_{OL}(s)$ and $\hat{\phi}_{CL}(s)$. Similarly, by defining $\hat{L}(s) \triangleq R^{1/2}L(s)R^{-1/2}$, we may work with $\hat{G}(s)$ and $\hat{L}(s)$ instead of $G(s)$ and $L(s)$. The conditions (3.6) and (3.7) of Theorem 3.1 are equivalent to $\underline{\sigma}(I + \hat{G}(s)) > 1$ and $\underline{\sigma}(I + \hat{G}(s)) \geq 1$, respectively. Condition (3.16) and Theorem 3.1 require that

$$\bar{\sigma}(\hat{L}^{-1}(s) - I) \leq 1 < \underline{\sigma}(I + \hat{G}(s)), \quad s \in \Omega_R \quad (\text{A.44})$$

and by Theorem 2.3 we conclude that $\hat{\phi}_{CL}(s)$ has no CRHP zeros. Alternatively condition (3.17) and Theorem 3.1 require that

$$\bar{\sigma}(\hat{L}^{-1}(s) - I) < 1 \leq \underline{\sigma}(I + \hat{G}(s)) \quad (\text{A.45})$$

which again by Theorem 2.3 means $\hat{\phi}_{CL}(s)$ has no CRHP zeros. Q.E.D.

Proof of Corollary 3.1: From Theorems 3.1 and 3.2 we know if $Q > 0$ then

$$\bar{\sigma}(R^{1/2}L^{-1}(s)R^{-1/2} - I) = \bar{\sigma}(L^{-1}(s) - I) \leq 1, \quad s \in \Omega_R \quad (\text{A.46})$$

to satisfy (3.16) when $L(s)$ and R are diagonal. If $\phi_{CL}(j\omega) \neq 0 \forall \omega$ then

$$\bar{\sigma}(L^{-1}(s) - I) < 1, \quad s \in \Omega_R \quad (\text{A.47})$$

to satisfy (3.17) when $L(s)$ and R are diagonal. The remainder of the proof of completely analogous to Corollary 2.1. Q.E.D.

Proof of Corollary 3.2: Only conditions (3.16) and (3.17) of Theorem 3.2 need to be verified for the $L(s)$ of (2.42) the rest are satisfied by assumption. Note that for $s \in \Omega_R$

$$\begin{aligned} \bar{\sigma}(R^{1/2}L^{-1}(s)R^{-1/2} - I) \\ = \bar{\sigma} \left(\begin{bmatrix} 0 & -R_1^{1/2}X(s)R_2^{-1/2} \\ 0 & 0 \end{bmatrix} \right) \quad \text{or} \\ \bar{\sigma} \left(\begin{bmatrix} 0 & 0 \\ -R_2^{1/2}X(s)R_1^{-1/2} & 0 \end{bmatrix} \right) \\ \leq \bar{\sigma}(X(s)) \max \{ \bar{\sigma}(R_1^{1/2})\bar{\sigma}(R_2^{-1/2}), \bar{\sigma}(R_2^{1/2})\bar{\sigma}(R_1^{-1/2}) \} \end{aligned}$$

Proof of Theorem 3.3: Conditions (3.38) and (3.39) and the Lyapunov stability criterion guarantee that condition (2.44) of Theorem 2.4 is satisfied. As in the proof of Theorem 3.2 we may work with $\hat{G}(s) = R^{1/2}G(s)R^{-1/2}$ and $\hat{L}(s) = R^{1/2}L(s)R^{-1/2}$ instead of $G(s)$ and $L(s)$. Condition (3.37) is simply condition (2.45) of Theorem 2.4 with $\hat{G}(s)$ replacing $G(s)$ and $\hat{L}(s)$ replacing $L(s)$ in (2.46) is simply (3.41). Thus by Theorem 2.4 the theorem is proved when (3.41) holds. When $Q > 0$ and (3.40) is satisfied, the strictness of the inequality (2.46) of Theorem 2.4 may be changed to $\geq$ and the $\geq$ of (2.45) to $>$ and Theorem 2.4 remains valid. Thus when (3.40) holds Theorem 3.1 is again proved. Q.E.D.

Proof of Theorem 3.4: Breaking the loop at point ① of Fig. 12 we have a loop transfer function matrix of

$$\begin{aligned} G_r(sI - A + G_f C)^{-1} [G_f C(Is - A)^{-1} B + B] \\ = G_r(sI - A)^{-1} B \triangleq G(s) \quad (\text{A.50}) \end{aligned}$$

so that

$$V \triangleq [v_1, v_2, \dots, v_n]; \quad V^H V = I \quad (\text{B.4})$$

$$\Sigma = \text{diag} [\sigma_1, \sigma_2, \dots, \sigma_n] \quad (\text{B.5})$$

and the columns of V and U are eigenvectors of $A^H A$ and $A A^H$, respectively. The minimum and maximum singular values denoted $\underline{\sigma}$ and $\bar{\sigma}$, respectively, are sometimes equivalently defined in terms of the spectral matrix norm $\|\cdot\|_2$ as

$$\bar{\sigma}(A) = \max_{\|x\|_2 \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \|A\|_2 \quad (\text{B.6})$$

and

$$\begin{aligned} \underline{\sigma}(A) &= \min_{\|x\|_2 \neq 0} \frac{\|Ax\|_2}{\|x\|_2} \\ &= \begin{cases} \|A^{-1}\|_2^{-1}, & \text{if } \det A \neq 0 \\ 0, & \text{if } \det A = 0. \end{cases} \quad (\text{B.7}) \end{aligned}$$

The minimum singular value $\underline{\sigma}(A)$ provides a measure of the nearness to singularity of the matrix A in the following

- [10] G. Zames, "On the input-output stability of time-varying nonlinear feedback systems—Part I: Condition using concepts of loop gain conicity and positivity," *IEEE Trans. Autom. Contr.*, vol. AC-11, pp. 228–238, Apr. 1966.
- [11] G. Zames, "On the input-output stability of time-varying nonlinear feedback systems—Part II: Conditions involving circles in the frequency plane and sector nonlinearities," *IEEE Trans. Autom.*
- [41] M. Athans, "The role and use of the stochastic linear-quadratic-Gaussian problem in control system design," *IEEE Trans. Automat. Contr.*, vol. AC-16, pp. 529–552, Dec. 1971.
- [42] J. B. Lewis, "Automotive engine control: As linear-quadratic approach," S.M. thesis, Laboratory for Information and Decision Systems, MIT, Cambridge, MA, Mar. 1980.
- [43] I. C. Doyle and G. Stein, "Multivariable feedback design: Con-

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The Design of Controllers for the Multivariable Robust Servomechanism Problem Using Parameter Optimization Methods

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Abstract—The problem of designing realistic multivariable controllers to solve the servomechanism problem is considered in this paper. Specifically, it is desired to find a controller for a plant to solve the robust servomechanism problem, so that closed-loop stability and asymptotic regulation occur, and also so that other desirable properties of the controlled system, such as fast response, low-interaction, integrity, tolerance to plant variations, etc., occur. The method of design is based on using state space methods via a two-stage process: 1) using theory, determine the existence of a solution and control structure required to solve the problem, and 2) using nonlinear programming methods, determine the unknown controller parameters so as to minimize a performance index for the system subject to certain constraint requirements. Numerous examples, varying from a single-input/single-output to a four-input/four-output system, are given to illustrate the design method, and the results obtained are compared with the results obtained by using other alternate design methods. In all cases, the controllers obtained have been highly competitive with controllers obtained by alternate design methods.

I. INTRODUCTION

ONE of the most basic problems occurring in control engineering is the design problem associated with finding realistic controllers to solve the multivariable robust servomechanism problem. In this type of problem, a mathematical model of the plant to be controlled is assumed to be known to the designer, and various constraints, specific to the particular plant being studied must

be considered in the problem formulation. In addition, decisions re tradeoffs between speed of response, integrity, tolerance to plant parameter variations, etc. must be made by the designer. These types of considerations imply that human judgment (and not mathematics) must ultimately be used in the final "shaping" of a satisfactory controller to solve the problem. The purpose of this paper is to develop a tool which can be used by the designer to give him guidance in the final shaping of such a controller. Hopefully, the tool will have the properties that the designer need not worry about any "intermediate" calculations required to determine the controller, but only about the final engineering properties of the controller itself, e.g., deciding on tradeoffs between speed of response and integrity of the controller say.

In the multivariable robust servomechanism problem to be considered, it is desired to find a controller so that:

1) the closed-loop system is asymptotically stable, with

a desired "fast" transient behavior,

2) asymptotic tracking and regulation occur for all reference input signals of a given class, for all disturbance signals of a given class and such that "low interaction" occurs,

3) properties 1) and 2) hold in the presence of changes in the dynamics of the plant; in addition constraints on integrity, controller gains, damping of the system, etc. are to be satisfied.

The method of design adopted in this paper to form such a controller is based on state space methods using a two-stage process and is in the spirit of [1], [4], [20], [21]

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